Lasers in Medical Science (2026) 41:219
https://doi.org/10.1007/s10103-026-05019-8

RESEARCH

Efficacy of high-intensity laser therapy versus ultrasound therapy in patients with knee osteoarthritis: a randomized controlled trial

Chantanee Ninlerd[1] · Nithithorn Benjakulsap[1] · Chutikarn Amornsupak[1] · Gonrada Phuangphay[1] · Rapeepat Narkbunnam[1] · Nichakorn Khomawut[1]

Received: 18 May 2026 / Accepted: 31 August 2026
© The Author(s) 2026

Abstract

Purpose High-intensity laser therapy (HILT) and ultrasound (US) are widely used for knee osteoarthritis (KOA), but comparative efficacy data remain scarce. This study compared HILT and US as exercise adjuncts in patients with KOA.

Methods In this single-center, assessor-blinded RCT, 66 adults with KOA were randomized 1:1 to HILT or US twice weekly for 6 weeks as exercise adjuncts. The primary outcome was WOMAC total score change from baseline to 12 weeks post-treatment (minimum important change [MIC] = 10 points, applied as an approximation). Secondary outcomes included VAS, OKS, KOOS, and EQ-5D-5L.

Results In ITT analysis (N = 66), the HILT group achieved a mean WOMAC reduction of 40.0 vs 8.2 points (adjusted between-group difference: -27.7 points, 95% CI: −39.0 to −16.5; $p < 0.001$, partial $\eta^2 = 0.277$). At 12 weeks post-treatment, greater improvements across all secondary outcomes were observed in exploratory analyses ($p \leq 0.012$). Furthermore, HILT maintained therapeutic effects up to 12 weeks post-treatment, whereas the US group experienced a gradual loss of post-treatment gains.

Conclusion HILT combined with exercise produced greater short-term improvements in WOMAC total score than US in patients with KOA, with exploratory findings suggesting consistent benefits in pain, function, and quality of life, and with benefits maintained up to 12 weeks after the last treatment session.

Keywords Knee osteoarthritis · Physical therapy modalities · High-intensity laser therapy · Ultrasound therapy · Photobiomodulation

Introduction

Knee osteoarthritis (KOA) is a prevalent degenerative joint disease causing substantial functional disability, with pharmacological treatments often limited by side effects and insufficient long-term efficacy [1, 2]. Effective nonpharmacological interventions are therefore needed to manage pain, stiffness, and functional limitations [3].

Current management guidelines emphasize multimodal approaches, with physical therapy modalities playing a central role alongside exercise [4]. Among these methods, therapeutic ultrasound (US) is widely utilized in rehabilitation. A recent systematic review by He et al. demonstrated that the US can significantly reduce pain and improve function in patients with KOA, although the quality of evidence varies [5]. Its therapeutic value is attributed to both its thermal effects (deep tissue heating) and nonthermal effects (acoustic streaming and cavitation), which aim to promote

✉ Nichakorn Khomawut
nichakorn.kho@gmail.com

Chantanee Ninlerd
numchantanee@gmail.com

Nithithorn Benjakulsap
nithithorn.b@gmail.com

Chutikarn Amornsupak
chutikarnt25@gmail.com

Gonrada Phuangphay
auai_18@hotmail.com

Rapeepat Narkbunnam
mai_parma@hotmail.com

[1] Department of Orthopaedic Surgery, Faculty of Medicine Siriraj Hospital, Mahidol University, Bangkok, Thailand

tissue healing and reduce local inflammation [6]. Despite heterogeneity in the evidence base, US remains one of the most widely used physical therapy modalities for KOA in clinical practice, including in Thailand, and its parameters have been identified as effective and safe [7, 8]. Its selection as the active comparator in this trial reflects its clinical prevalence, enabling findings to directly inform the pragmatic decision faced by clinicians choosing between these two modalities in routine care.

High-intensity laser therapy (HILT) has recently emerged as a promising alternative. Unlike low-level laser therapy, HILT utilizes high-power lasers (typically > 500 mW) to deliver energy to deeper tissues via photobiomodulation [9]. These mechanisms involve the absorption of light energy by cellular chromophores, enhancing mitochondrial ATP production, cellular metabolism, and tissue oxygenation [10]. The evidence for HILT is growing; a network meta-analysis by Wu et al. revealed that HILT had the highest probability of being the most effective physical agent for pain reduction among the evaluated interventions [11].

Despite this promising evidence, direct head-to-head comparisons between HILT and established modalities such as ultrasound remain limited. While studies have favored HILT over placebo [12], there is a critical knowledge gap regarding how HILT compares directly to ultrasound when both are used in a standard clinical setting. Since both modalities are commonly available but utilize distinct physiological mechanisms, determining which modality offers greater clinical benefit is essential for optimizing clinical decision-making.

Therefore, this study aimed to compare the efficacy of HILT with that of ultrasound therapy as an adjunct to an exercise program in patients with knee osteoarthritis. We hypothesized that HILT would produce greater therapeutic improvements compared with ultrasound therapy.

Methods

Study design

We conducted a single-center, parallel-group, assessor-blinded, randomized controlled superiority trial with a 1:1 allocation ratio. The study protocol was approved by the Human Research Protection Unit, Faculty of Medicine Siriraj Hospital, Mahidol University (COA no. Si1042/2021). The participants were enrolled between September 2022 and September 2025 at the Department of Orthopaedic Surgery, Faculty of Medicine Siriraj Hospital, Mahidol University. This trial was designed and reported in accordance with the CONSORT 2010 statement. All participants provided written informed consent before enrollment. The study was retrospectively registered with the Thai Clinical Trial Registry (TCTR20251021008; registered 21 October 2025), as registration was not completed prior to recruitment commencement owing to a procedural oversight by the research team. A written SAP was finalized prior to data analysis; no changes to the analytical approach were made after data inspection. The outcomes, assessment time points, and eligibility criteria registered in the TCTR are fully consistent with those reported herein and with the ethics-approved protocol. Outcomes were assessed at baseline (T0), after 6 sessions (T1), immediately after completing treatment (T2), and 12 weeks after the last session (T3). The registered TCTR protocol is fully consistent with the ethics-approved protocol (COA no. Si1042/2021) and all outcomes reported herein, with no changes to methods, outcomes, or procedures after trial commencement.

Participants

Participants were recruited and screened for eligibility by a single orthopaedic surgeon at orthopaedic clinic of Siriraj hospital. The inclusion criteria were as follows: (1) knee OA diagnosis by ACR clinical classification criteria [13]; (2) Kellgren-Lawrence grade II–III; (3) chronic knee pain ≥ 3 months; (4) VAS pain score > 3; (5) BMI < 35 kg/m^2; and (6) ability to provide informed consent. For participants with bilateral KOA, the more symptomatic knee defined as the knee with greater pain severity at baseline was designated as the index knee for all outcome assessments and remained consistent throughout the study. Both knees received the allocated physical therapy modality at each treatment session; however, all outcome measures were completed with specific reference to the index knee only, and participants were explicitly instructed accordingly. Participants were additionally instructed to refrain from receiving any additional physiotherapy or physical modality treatments targeting the knee outside the study protocol for the duration of the intervention period. The exclusion criteria included (1) knee trauma or metal implants; (2) physical therapy within the previous 3 months; (3) intra-articular corticosteroid injections within 6 months; (4) inflammatory arthritis disease; and (5) neuromuscular disease. Participants who missed more than three sessions were withdrawn.

Treatment protocol

Both groups attended twice-weekly physiotherapy sessions for 6 weeks (12 sessions total), each lasting approximately 30 min. All participants were placed in a supine position with the knee flexed to 30°. For participants with bilateral KOA, both knees received the allocated physical therapy

modality at each treatment session, consistent with standard clinical practice.

HILT was delivered via a Nd:YAG laser (BTL-6000, BTL Industries, UK; 1064 nm) in two sequential phases. The analgesic phase was delivered in pulsed mode (frequency: 25 Hz; peak power: 10 W; energy density: 12 J/cm^2; treatment area: 25 cm^2; total energy: 300 J; duration: 2 min per area), applied in a spiral scanning motion over the medial and lateral aspects of the knee. The biostimulation phase was delivered in continuous mode (power: 12 W; energy density: 80 J/cm^2; treatment area: 25 cm^2; total energy: 2,000 J; duration: approximately 3 min per area), applied via transverse scanning motion over the medial and lateral aspects of the knee, with emphasis on the most tender areas. The total treatment time was approximately 12 min per session. These parameters were selected based on protocols reported in prior RCTs demonstrating clinical efficacy in KOA [14, 15] and are consistent with the dosimetry range identified as most effective in a network meta-analysis of physical agents for KOA [11]. Patients were asked to wear protective eyewear during the HILT session.

US was administered via an INTELECT® MOBILE system (Model 2776, Chattanooga) in continuous mode (1 MHz, 1.0 W/cm^2, 100% duty cycle), which was applied in a circular motion over the medial and lateral knees for 5 min per area (total: ~ 10 min/session) [7, 8].

All participants concurrently performed an identical standardized home exercise program (3 sets of 10–15 repetitions daily; hamstring/gastrocnemius stretching, straight leg raises, long-arc quadriceps, hip abduction, hip extension) at fixed intensity with no progression, ensuring between-group differences reflect the allocated modality. Technique was reviewed at each session and adherence monitored via self-reported diary and verbal confirmation.

Outcome measurements

All outcome measures were assessed with reference to the index knee only. For participants with bilateral KOA, participants were explicitly instructed to respond to all questionnaires based on their more symptomatic knee, which had been designated as the index knee at baseline screening.

Primary outcome measure

The primary outcome was the change in the Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) total score from T0 to T3. We used the modified Thai version (WOMAC VA 3.01), which is valid and reliable for patients with KOA and consists of 3 subscales (pain, stiffness and physical function, 0–10 per item; total range 0–220). Lower scores indicate better clinical status (less pain and disability). As a formally validated MIC for the Thai WOMAC 3.1 raw score (0–220) has not been established, a threshold of 10 points was applied as a reference value informed by prior literature [16], while acknowledging that this value was originally derived from a normalized WOMAC (0–100) in a post-TKA population and represents an approximation in the present context. No normalization or transformation was applied; all scores reported are raw values on this 0–220 scale.

Secondary outcome measures

- Pain severity was assessed using the visual analog scale (VAS; 0–10 cm).
- The Oxford Knee Score (OKS; 12 items; 0–48; higher = better function) was used [17].
- The Thai KOOS was used to assess functional outcomes (42 items; 5 subscales; higher scores = better function) [18, 19].
- The EQ-5D-5L utility index (0–1) and EQ-VAS (0–100) assess health status; both Thai versions are validated [20, 21]. No changes to the primary or secondary outcomes were made after the trial commenced.

Sample size calculation

The sample size was estimated based on WOMAC total score data from Yıldırım et al. (2015) [22], in which patients receiving ultrasound therapy combined with exercise achieved a post-treatment WOMAC score of 36.08 ± 14.045 points. Assuming a clinically meaningful additional reduction of 10 points in the HILT group informed by the minimum important change threshold reported by Clement et al. (2018) [16] a minimum of 32 participants per group was required ($\alpha = 0.05$, 80% power; $d \approx 0.71$). Allowing for 10% dropout, the recruitment target was set at 35 per group (N = 70). The reference SD was derived from a Likert-format WOMAC [22] rather than the Thai WOMAC 3.1 VAS format (0–220) used here [23]; power calculation parameters should be interpreted accordingly. Sixty-six participants were enrolled, exceeding the a priori minimum of 64 (32 per group) required to detect the anticipated effect, although marginally below the dropout-adjusted recruitment target of 70. All 66 randomized participants were included in the intention-to-treat analysis. This shortfall, together with related power calculation imprecisions, is acknowledged as a limitation. No interim analyses were planned or conducted and no stopping guidelines were applied.

Randomization and blinding

Eligible participants were randomly assigned (1:1) without stratification to either the HILT group or the US group. Randomization was performed via computer-generated permuted block randomization (block size 4) by an independent researcher. Allocation concealment was ensured via sequentially numbered opaque sealed envelopes. Owing to the nature of the interventions, treating physiotherapists and participants could not be blinded. The enrolling physician, outcome assessors, and data analysts were blinded to group assignment and participants were instructed not to disclose their treatment. No accidental unblinding was reported.

Adverse events (skin redness, burning, pain exacerbation) were actively solicited before and after every treatment session. To minimize cointerventions, participants were instructed to discontinue analgesics, avoid injections, and refrain from additional physiotherapy throughout the intervention period; compliance was verbally confirmed at each session.

Statistical analysis

The data were analyzed via SPSS version 30 (IBM Corp., Armonk, NY, USA). Continuous variables are presented as mean ± SD and categorical variables as frequencies and percentages. Normality was assessed via the Shapiro–Wilk test. Baseline characteristics were compared using independent samples t-test and chi-square tests. An intention-to-treat (ITT) analysis was conducted including all 66 randomized participants; missing outcome data for the four withdrawn participants were imputed using last observation carried forward (LOCF), with baseline values carried forward to all subsequent time points. A per-protocol sensitivity analysis (N = 62) was also conducted; as the withdrawn participants had no post-baseline data, this is numerically equivalent to a complete-case analysis. Since significant baseline differences were observed in EQ-VAS ($p = 0.044$), ANCOVA with baseline scores as covariates was applied to all between-group comparisons at T1, T2, and T3. Effect sizes are reported as partial eta squared ($\eta^2 p$). Within-group changes were evaluated via repeated-measures ANOVA with Bonferroni correction. The primary confirmatory analysis was the between-group difference in WOMAC at T3 ($\alpha = 0.05$); WOMAC comparisons at T1/T2 and all secondary outcomes were prespecified as exploratory, and p-values should be interpreted without multiplicity adjustment. All tests were two-tailed with significance set at $p < 0.05$. All analyses were conducted according to the originally assigned groups, with no crossover or group reassignment throughout the study period.

Results

A total of 73 patients were screened; 7 were excluded (4 did not meet the inclusion criteria and 3 declined). The remaining 66 patients were randomized to the HILT group (HILT + EX, n = 33) or the US group (US + EX, n = 33). Four participants were withdrawn during treatment (2 per group) because more than three sessions were missing, leaving 66 participants (33 per group) for the ITT analysis (Fig. 1). All 66 participants contributed outcome data at T1, T2, and T3; LOCF imputation was applied to the four withdrawn participants using baseline values as the carried-forward observation for all subsequent time points.

The baseline demographic and clinical characteristics are summarized in Table 1. The groups were well matched, with no significant differences in any demographic variables ($p > 0.05$), although the study population was predominantly female (100% in the US group vs 87.9% in the HILT group).

A significant baseline difference was found in EQ-VAS ($p = 0.044$); all between-group comparisons were therefore adjusted via ANCOVA with baseline scores as covariates.

All 62 protocol completers attended all 12 treatment sessions without deviation (31 per group). Exercise adherence was monitored via diary and verbal confirmation; compliance rates were not formally quantified.

Primary outcome

At T3, the HILT group demonstrated significantly greater improvement in the WOMAC total score than did the US group (adjusted difference = −27.7 points, 95% CI: −39.0 to −16.5; $p < 0.001$, partial $\eta^2 = 0.277$), with a mean reduction of 40.0 points versus 8.2 points in the US group, substantially exceeding the approximate MIC threshold and suggesting a clinically meaningful treatment effect (Table 2). Significant between-group differences favoring HILT were already present at T1 (adjusted difference = −16.1 points, 95% CI: −26.1 to −6.0; $p = 0.002$), indicating faster onset as early as 3 weeks. During the maintenance phase, the HILT group maintained WOMAC improvements with no significant deterioration from T2 to T3, whereas the US group experienced a gradual loss of post-treatment gains (60.3 at T2 to 68.7 at T3).

Secondary outcome

At T3 (12 weeks post-treatment), in exploratory analyses, the HILT group showed significantly greater improvements than the US group across all secondary outcomes (Table 2): VAS (adjusted difference = −2.0 points, 95%

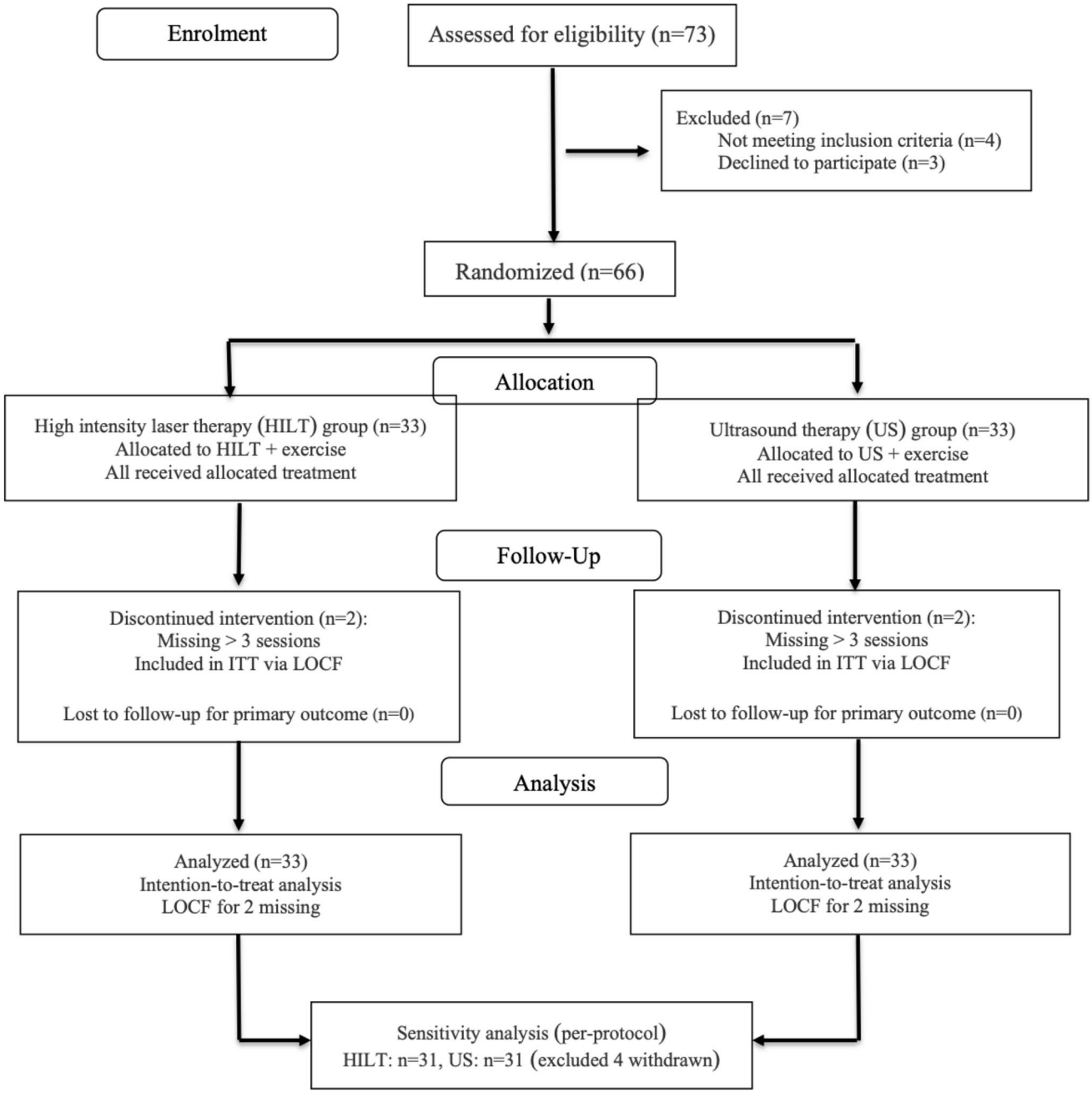


Fig. 1 CONSORT 2010 Flow diagram. LOCF = last observation carried forward

CI: −2.5 to −1.6; p < 0.001), OKS (+4.9 points, 95% CI: 2.6 to 7.2; p < 0.001), KOOS (+11.0 points, 95% CI: 6.7 to 15.3; p < 0.001), EQ-VAS (+6.7 points, 95% CI: 1.5 to 12.0; p = 0.012), and EQ-Utility (+0.074, 95% CI: 0.032 to 0.116; p < 0.001). These findings should be interpreted with caution given the exploratory nature of secondary outcomes and the absence of multiplicity correction. Between-group differences were not statistically significant across all outcomes at T2, with OKS (p = 0.057), EQ-VAS (p = 0.204), and EQ-Utility (p = 0.086) not reaching significance at that time point. During the maintenance phase, the HILT group preserved functional and quality of life gains with no significant decline in OKS, KOOS, or EQ-Utility, whereas the US group experienced a gradual loss of post-treatment gains across most secondary outcomes. No adverse events including skin reactions, burns, pain flares, or treatment-related discomfort were reported in either group throughout the intervention period.

Table 1 Baseline demographic and clinical characteristics of the participants

	US group (n = 33)	HILT group (n = 33)	p Value
Age(years)	64.8 ± 6.3	64.0 ± 9.9	0.691
Gender			0.114
Male	0 (0%)	4 (12.1%)	
Female	33 (100%)	29 (87.9%)	
Weight(kg)	60.6 ± 10.5	63.6 ± 10.9	0.251
Height(cm)	157.6 ± 5.5	159.8 ± 7.1	0.156
BMI (kg/m^2)	24.3 ± 3.7	24.9 ± 3.7	0.507
Affected side			0.423
Right	8 (24.2%)	6 (18.2%)	
Left	4 (12.1%)	8 (24.2%)	
Bilateral	21 (63.6%)	19 (57.6%)	

BMI: body mass index

Discussion

This RCT aimed to compare the clinical efficacy of HILT and US in patients with KOA. Our findings indicate that HILT demonstrated significantly greater improvement than US in the primary outcome, with a between-group difference in the WOMAC score of 27.7 points, which substantially exceeded the approximate MIC threshold and is consistent with a clinically meaningful treatment effect; exploratory analyses additionally suggested greater improvements across secondary outcomes. Furthermore, HILT maintained its treatment effect up to 12 weeks after the last treatment session, whereas post-treatment gains in the US group gradually diminished once treatment ceased.

Table 2 Between-group and within-group comparisons of all outcomes across time points

	Time interval	US Group(n = 33) mean ± SD	HILT Group(n = 33) mean ± SD	Adjusted Difference (95% CI) *	p value[1]	Partial Eta Squared
WOMAC total score	T0	76.9 ± 28.2	88.2 ± 33.9	N/A	0.182[a]	N/A
	T1	65.4 ± 23.2	57.2 ± 37.2	−16.1 (−26.1 – −6.0)	0.002[b]	0.139
	T2	60.3 ± 23.5	47.3 ± 38.5	−19.6 (−32.2 – −7.1)	0.002[b]	0.134
	T3	68.7 ± 22.8	48.2 ± 38.0	−27.7 (−39.0 – −16.5)	<0.001[b]	0.277
	p value[2]	<0.001 [m, n, r]	<0.001 [m, n, o, p, q]			
VAS Score	T0	5.1 ± 1.7	5.8 ± 1.4	N/A	0.055[a]	N/A
	T1	3.6 ± 1.3	2.7 ± 1.1	−1.4 (−1.8 – −1.0)	<0.001[b]	0.418
	T2	2.7 ± 1.2	1.7 ± 1.1	−1.3 (−1.8 – −0.9)	<0.001[b]	0.359
	T3	4.0 ± 1.4	2.5 ± 1.2	−2.0 (−2.5 – −1.6)	<0.001[b]	0.561
	p value[2]	<0.001 [m, n, o, p, q, r]	<0.001 [m, n, o, p, r]			
Oxford Knee score	T0	29.7 ± 6.9	27.2 ± 7.3	N/A	0.162a	N/A
	T1	32.5 ± 6.9	33.2 ± 6.7	2.6 (0.4–4.7)	0.022[b]	0.080
	T2	34.5 ± 6.6	35.1 ± 7.5	2.4 (−0.1 – 4.8)	0.057[b]	0.056
	T3	32.0 ± 5.9	35.2 ± 7.2	4.9 (2.6–7.2)	<0.001[b]	0.228
	p value[2]	<0.001 [m, n, o, p, r]	<0.001 [m, n, o, p, q]			
KOOS	T0	56.3 ± 16.4	50.6 ± 13.4	N/A	0.119[a]	N/A
	T1	61.4 ± 13.3	63.1 ± 13.3	5.8 (1.8–9.7)	0.005[b]	0.119
	T2	66.6 ± 14.7	69.3 ± 14.8	7.1 (2.4–11.9)	0.004[b]	0.124
	T3	60.7 ± 13.5	67.7 ± 13.5	11.0 (6.7–15.3)	<0.001[b]	0.290
	p value[2]	<0.001 [m, n, o, p, r]	<0.001 [m, n, o, p, q]			
EQ VAS	T0	71.3 ± 14.2	63.9 ± 14.3	N/A	0.044[a]	N/A
	T1	73.9 ± 12.4	74.2 ± 14.7	5.3 (0.8–9.7)	0.022[b]	0.080
	T2	78.0 ± 17.1	81.0 ± 11.8	4.6 (−2.6 – 11.7)	0.204[b]	0.025
	T3	74.8 ± 10.9	79.9 ± 11.5	6.7 (1.5–12.0)	0.012[b]	0.096
	p value[2]	0.053	<0.001 [m, n, o, p, q]			
EQ Utility	T0	0.75 ± 0.2	0.68 ± 0.2	N/A	0.107[a]	N/A
	T1	0.81 ± 0.1	0.81 ± 0.1	0.041 (0.003–0.079)	0.033[b]	0.070
	T2	0.85 ± 0.10	0.85 ± 0.16	0.040 (−0.006– 0.086)	0.086[b]	0.046
	T3	0.82 ± 0.10	0.85 ± 0.15	0.074 (0.032–0.116)	<0.001[b]	0.165
	p value[2]	<0.001 [m, n, r]	<0.001 [m, n, o]			

a: Independent t-test, b: Analysis of covariance (covariate by baseline), P value[1]: comparison between groups, P value [2]: comparison within groups, m: T0 vs T1 significant, n: T0 vs T2 significant, o: T0 vs T3 significant, p: T1 vs T2 significant, q: T1 vs T3 significant, r: T2 vs T3 significant (Significant = $p < 0.05$; absence of a letter indicates the comparison was conducted but did not reach statistical significance ($p \geq 0.05$))

- For WOMAC and VAS, negative values indicate greater clinical improvement (lower symptom severity) in the HILT group. For Oxford Knee Score and KOOS, positive values indicate greater clinical improvement (better joint function) in the HILT group. For EQ-VAS and EQ-Utility, positive values indicate better health status in the HILT group

- Within-group p-values (p-value 2) are presented for descriptive purposes only. Primary inferential conclusions are based on between-group adjusted differences and 95% confidence intervals

Secondary outcomes are exploratory; p-values should be interpreted without multiplicity adjustment. The significant improvements in pain relief (VAS) and functional capacity (WOMAC, KOOS, and Oxford Knee score) observed in the HILT group align with recent findings highlighting the advanced therapeutic effects of high-intensity photobiomodulation. Previous studies by Kheshie et al. (2014) [14] and Angelova et al. (2016) [15] have similarly reported that HILT yields greater pain reduction and functional improvement in KOA patients compared with standard physical therapy modalities.

Our results align with recent network meta-analyses highlighting HILT's greater efficacy, ranking it highest for pain reduction in patients with knee osteoarthritis [11]. While previous studies established HILT's greater effectiveness over exercise alone [24] or placebo [12], the present trial addresses a critical knowledge gap by providing a direct, head-to-head comparison with an active comparator (continuous ultrasound). By isolating the specific additional value of HILT against a standard-of-care modality, our observed treatment effect falls within the upper range of pooled estimates [11], further supporting its comparative effectiveness over ultrasound in this patient population. The magnitude of the between-group WOMAC difference (27.7 points, raw scale 0–220) should be interpreted contextually; absolute differences appear numerically larger on the Thai WOMAC 3.1 raw scale than on normalized versions used in prior studies, and the gradual loss of post-treatment gains in the US group during the maintenance phase further widened the gap at T3. While consistent with prior HILT evidence [11, 14, 15], potential contributions from expectation bias cannot be entirely excluded, as acknowledged in the Limitations section.

As this trial was designed as a pragmatic active comparator study rather than a mechanistic investigation, the observed between-group differences reflect comparative effectiveness as delivered in clinical practice and cannot be attributed solely to the specific physiological effects of laser energy. The maintained clinical difference observed between groups may be explained by differences in mechanisms of action and energy dosimetry, though these explanations remain speculative as no mechanistic outcomes were measured in this study. Unlike ultrasound, which relies primarily on superficial thermal effects and acoustic streaming [6, 25], HILT utilizes high peak power enabling deep articular tissue penetration without causing epidermal damage [9]. Based on evidence from prior laboratory and experimental studies, it has been hypothesized that this energy is absorbed by cellular chromophores, stimulating mitochondrial oxidative phosphorylation and ATP production to accelerate tissue repair, alongside modulation of proinflammatory cytokines [10, 26]. The substantially greater total energy delivered per session in the HILT protocol (2,000 J) compared with the standard ultrasound protocol may have contributed to the magnitude of the observed comparative effectiveness within the study period, though this cannot be confirmed without direct physiological measurement.

The short-term comparative effectiveness of HILT observed in this trial suggests it may warrant consideration as an adjunct nonpharmacological option in KOA management; whether short-term benefits extend to long-term outcomes or reduce invasive intervention rates requires confirmation in longer follow-up trials.

Several methodological strengths merit consideration. The head-to-head design against an active comparator, rather than a placebo, allowed direct isolation of HILT's specific therapeutic value over a standard-of-care modality. Standardization of the home exercise protocol across both groups further minimized cointervention bias. Additionally, the 12-week post-treatment follow-up provided preliminary evidence of short-term treatment maintenance, addressing a gap commonly left by shorter evaluations. However, the generalizability of these findings is limited by the predominantly female sample (HILT: 87.9%, US: 100% female), and the absence of male participants in the US group precludes assessment of potential sex-by-treatment interactions.

This study has several limitations. First, the 12-week follow-up did not confirm long-term durability of outcomes. Second, blinding of therapists and participants was not feasible, potentially introducing performance and expectation bias; no treatment credibility assessments were conducted to quantify this influence. Detection bias was minimized through blinding of the enrolling physician, outcome assessors, and data analysts, and the active comparator design partially mitigated differential expectation effects. The absence of a sham control group further means that specific physiological effects of HILT cannot be fully distinguished from non-specific effects including patient expectations, therapist interaction, and the concurrent exercise program. These factors, alongside the scoring characteristics of the Thai WOMAC 3.1 raw scale (0–220) and the gradual loss of post-treatment gains in the US group during the maintenance phase, should be considered when interpreting the magnitude of the between-group difference. Future trials should consider incorporating treatment credibility scales and sham-controlled arms. Third, a significant baseline difference in EQ-VAS ($p=0.044$) was observed despite randomization, likely reflecting chance imbalance in a moderate sample without stratification; this was addressed via ANCOVA with baseline covariates, though residual confounding cannot be excluded. Kellgren-Lawrence grade, used only as an eligibility criterion (II–III) and not formally recorded as a study variable, could not be compared between groups; this narrow eligible range may partially

mitigate the potential impact of this unmeasured imbalance. Fourth, the study population was predominantly female (HILT: 87.9%, US: 100%), limiting generalizability to male patients with KOA. The absence of male participants in the US group further precludes assessment of sex-related differential treatment responses, and findings should be interpreted with caution regarding potential sex-by-treatment interactions. Fifth, this trial was registered retrospectively due to a procedural oversight; however, the protocol and all outcomes were ethics-approved prior to recruitment (COA no. Si1042/2021, December 2021), a written SAP was finalized before data analysis, and the registered protocol is fully consistent with outcomes reported herein. Sixth, the achieved sample size (66 randomized; 62 per-protocol) exceeded the a priori minimum required for adequate power (64) but fell short of the dropout-adjusted target of 70, and this shortfall is acknowledged as a limitation rather than offset by the magnitude of the observed effect. Missing outcome data for the four withdrawn participants were imputed using LOCF, whereby baseline values were carried forward for all subsequent time points; while this conservative approach assumes no improvement in those who withdrew, it enabled a full intention-to-treat analysis including all 66 randomized participants. The MIC and reference SD used in the power calculation were derived from a different WOMAC scoring version and clinical context, representing acknowledged imprecisions. Additionally, cointervention and home exercise compliance were monitored via verbal confirmation and self-reported diary respectively, but neither was formally quantified, limiting verifiability of protocol adherence. Use of separate ANCOVA models at each time point rather than a linear mixed-effects model may increase multiplicity risk; future trials should consider linear mixed-effects modelling. Seventh, for participants with bilateral KOA, both knees received treatment and outcomes were attributed to the more symptomatic index knee; contralateral carry-over effects cannot be entirely excluded. Eighth, the ultrasound parameters, while evidence-based, may not represent optimized comparator dosimetry. Finally, the reliance on patient-reported outcomes without objective functional performance measures limits the clinical interpretability of functional improvement claims. Future trials should incorporate performance-based outcomes alongside patient-reported measures for a more comprehensive evaluation. Furthermore, the speed of onset of treatment effect was not prespecified as a primary or secondary outcome; the between-group difference observed at T1 provides preliminary exploratory evidence of faster symptomatic improvement with HILT but should be interpreted with caution.

Future research should incorporate extended follow-up periods to evaluate the medium- and long-term durability of HILT beyond the short-term maintenance observed in this study and recruit more gender-balanced samples to increase generalizability. Subsequent trials should also integrate objective functional performance tests and structural or biological assessments such as MRI or synovial fluid analysis to directly validate the proposed photobiomodulation and cartilage repair mechanisms.

Conclusion

In this single-center RCT, HILT plus exercise produced greater short-term improvements in patient-reported pain and function than ultrasound plus exercise in patients with KOA, with benefits maintained up to 12 weeks post-treatment. However, these findings require confirmation in prospectively registered, adequately powered, blinded or sham-controlled trials incorporating objective functional outcomes, longer follow-up periods, and more sex-balanced samples before definitive conclusions regarding the role of high-dosimetry HILT (2,000 J per session) in KOA management can be drawn. These findings are most applicable to middle-aged and older female patients with KOA.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10103-026-05019-8.

Acknowledgements The authors would like to thank all participants in this research, the Routine to Research Unit (Faculty of Medicine Siriraj Hospital, Mahidol University), and Mr. Arnuphap Tanasakampai for consultation in this research.

Author contribution C.N.: project administration, conceptualization, methodology, data curation, software, writing and editing, N.B., C.A., and G.P.: methodology and data curation, R.N.: conceptualization, supervision, writing and editing, N.K.: conceptualization, methodology, data curation, writing and editing. All authors have read and agreed to the published version of the manuscript.

Funding This work was supported by the Routine to Research Unit, Faculty of Medicine Siriraj Hospital, Mahidol University [grant number R016535023].

Data availability No datasets were generated or analysed during the current study.

Declarations

Ethics approval The questionnaire and methodology for this study was approved by the Human Research Protection Unit, Faculty of Medicine Siriraj Hospital, Mahidol University (COA no. Si1042/2021) on December 29, 2021.

Consent of participants Written informed consent was obtained from all participants prior to enrollment.

Trial Registration Thai Clinical Trials Registry, TCTR20251021008, registered 21 October 2025 (retrospectively registered).

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

References

1. Sinusas K (2012) Osteoarthritis: diagnosis and treatment. Am Fam Physician 85:49–56
2. Da Costa BR, Reichenbach S, Keller N, Nüesch E, Jüni P (2017) Effectiveness of non-steroidal anti-inflammatory drugs for the treatment of pain in knee and hip osteoarthritis: a network meta-analysis. Lancet 390:e21–e33. https://doi.org/10.1016/S0140-6736(17)31744-0
3. Plotnikoff R, Karunamuni N, Lytvyak E et al (2015) Osteoarthritis prevalence and modifiable factors: a population study. BMC Public Health 15:1195. https://doi.org/10.1186/s12889-015-2529-0
4. Bannuru RR, Osani MC, Vaysbrot EE et al (2019) OARSI guidelines for the non-surgical management of knee, hip, and polyarticular osteoarthritis. Osteoarthritis Cartilage 27:1578–1589. https://doi.org/10.1016/j.joca.2019.06.011
5. He Z, Fan J, Wang Z, Li S (2024) Effects of therapeutic ultrasound in patients with knee osteoarthritis: a systematic review and meta-analysis. Heliyon 10:e30874. https://doi.org/10.1016/j.heliyon.2024.e30874
6. Baker KG, Robertson VJ, Duck FA (2001) A review of therapeutic ultrasound: biophysical effects. Phys Ther 81:1351–1358
7. Özgönenel L, Aytekin E, Durmuşoğlu G (2009) A double-blind trial of clinical effects of therapeutic ultrasound in knee osteoarthritis. Ultrasound Med Biol 35(1):44–49. https://doi.org/10.1016/j.ultrasmedbio.2008.07.009
8. Dantas LO, Osani MC, Bannuru RR (2021) Therapeutic ultrasound for knee osteoarthritis: a systematic review and meta-analysis with grade quality assessment. Braz J Phys Ther 25(6):688–697. https://doi.org/10.1016/j.bjpt.2021.07.003
9. Berlien HP, Müller GJ (eds) (2003) Applied Laser Medicine. Springer, Berlin, Heidelberg
10. Monici M, Cialdai F, Fusi F, Romano G, Pratesi R (2009) Effects of pulsed Nd:YAG laser at molecular and cellular level: a study on the basis of Hilterapia. Energy for Health 3:26–33
11. Wu M, Luan L, Pranata A, Witchalls J, Adams R, Bousie J, Han J (2022) Is high intensity laser therapy more effective than other physical therapy modalities for treating knee osteoarthritis? A systematic review and network meta-analysis. Front Med 9:956188. https://doi.org/10.3389/fmed.2022.956188
12. Ekici B, Ordahan B (2023) Evaluation of the effect of high-intensity laser therapy (HILT) on function, muscle strength, range of motion, pain level, and femoral cartilage thickness in knee osteoarthritis: randomized controlled study. Lasers Med Sci 38:218. https://doi.org/10.1007/s10103-023-03887-y
13. Altman R, Asch E, Bloch D et al (1986) Development of criteria for the classification and reporting of osteoarthritis: classification of osteoarthritis of the knee. Arthritis Rheum 29:1039–1049. https://doi.org/10.1002/art.1780290816
14. Kheshie AR, Alayat MSM, Ali MME (2014) High-intensity versus low-level laser therapy in the treatment of patients with knee osteoarthritis: a randomized controlled trial. Lasers Med Sci 29:1371–1376. https://doi.org/10.1007/s10103-014-1529-0
15. Angelova A, Ilieva EM (2016) Effectiveness of high intensity laser therapy for reduction of pain in knee osteoarthritis. Pain Res Manag 2016:9163618. https://doi.org/10.1155/2016/9163618
16. Clement ND, Bardgett M, Weir D, Holland J, Gerrand C, Deehan DJ (2018) What is the minimum clinically important difference for the WOMAC index after TKA? Clin Orthop Relat Res 476:2005–2014
17. Charoencholvanich K, Pongcharoen B (2005) Oxford Knee Score and SF-36: translation and reliability for use with total knee arthroplasty patients in Thailand. J Med Assoc Thai 88:1194–1202
18. Chaipinyo K (2009) Test-retest reliability and construct validity of Thai version of Knee Osteoarthritis Outcome Score (KOOS). Thai J Phys Ther 31:67–76
19. Roos EM, Roos HP, Lohmander LS, Ekdahl C, Beynnon BD (1998) Knee injury and osteoarthritis outcome score (KOOS) — development of a self-administered outcome measure. J Orthop Sports Phys Ther 28:88–96. https://doi.org/10.2519/jospt.1998.28.2.88
20. Pattanaphesaj J (2014) Health-related quality of life measure (EQ-5D-5L): measurement property testing and its preference-based score in Thai population. Doctoral dissertation, Mahidol University, Nakorn Pathom
21. Herdman M, Gudex C, Lloyd A et al (2011) Development and preliminary testing of the new five-level version of EQ-5D (EQ-5D-5L). Qual Life Res 20:1727–1736. https://doi.org/10.1007/s11136-011-9903-x
22. Yıldırım MA, Uçar D, Öneş K (2015) Comparison of therapeutic duration of therapeutic ultrasound in patients with knee osteoarthritis. J Phys Ther Sci 27:3667–3670
23. Kuptniratsaikul V, Rattanachaiyanont M (2007) Validation of a modified Thai version of the Western Ontario and McMaster (WOMAC) Osteoarthritis Index for knee osteoarthritis. Clin Rheumatol 26:1641–1645. https://doi.org/10.1007/s10067-007-0560-y
24. Khalilizad M, Hosseinzade D, Marzban Abbas Abadi M (2024) Efficacy of high-intensity and low-level laser therapy combined with exercise therapy on pain and function in knee osteoarthritis: a systematic review and network meta-analysis. J Lasers Med Sci 15:e34
25. Rutjes AW, Nüesch E, Sterchi R, Jüni P (2010) Therapeutic ultrasound for osteoarthritis of the knee or hip. Cochrane Database Syst Rev 1:CD003132. https://doi.org/10.1002/14651858.CD003132.pub2
26. Zhang Y, Ji Q (2023) Current advances of photobiomodulation therapy in treating knee osteoarthritis. Front Cell Dev Biol 11:1286025. https://doi.org/10.3389/fcell.2023.1286025

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.